

Sriharsha Annamaneni

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ML engineer with 7+ years shipping production AI across healthcare, insurance, and automotive. The last two years have focused on LLM systems: RAG platforms, agent orchestration, model fine-tuning, and the serving infrastructure behind them. Prior work at Bosch covered computer vision and applied NLP. Current focus areas are enterprise RAG, agentic workflows with LangGraph and MCP, and training and serving optimization for production LLM deployments.

Work Experience

Senior Consultant

May 2024 – Present

Fractal, Bengaluru

- **Health Insurance SOP Assistant.** Built and scaled an enterprise RAG platform now used by 5K+ daily users. Hybrid retrieval (vector + keyword + department-scoped metadata) on a tuned HNSW index, an event-driven ingestion pipeline on Azure Functions + Service Bus, and role-based access tied to Azure AD group resolution. Landed at 90% accuracy with sub-second latency. Supported a program credited with \$11M+ in operational savings.
- Response quality improved 40% over the initial baseline during UAT. The jump came from implementing the DSPy evaluator-optimizer pattern, wiring SME feedback into a refinement loop, using GEPA for prompt/program optimization, and running Best-of-N sampling for the most sensitive query types.
- Wrote a distributed token caching and rate-limiting service in FastAPI + Redis with Lua scripts for atomicity. This killed the token-refresh thundering herd we were seeing and made shared OpenAI TPM quotas work cleanly across parallel workers.
- Set up the document normalization layer (grammar cleanup, abbreviation expansion, HTML/PDF parsing) which measurably improved chunk quality and retrieval relevance, and cut refresh cycles by roughly 10x.
- **Text2SQL Chatbot.** Led the build of an enterprise Text2SQL chatbot on LangChain + LangGraph with a multi-agent layout (router, clarity, text2SQL, output). Azure AI Search handled retrieval with guardrails for PII and hallucination. Shipped to AKS behind a Streamlit frontend; SQL generation accuracy improved 27% vs. the previous prompt-only approach.
- **Appeals Verification and Enrollment Automation.** Built a ReAct-style reasoning pipeline in LangGraph for appeals decisions, with an MCP server in front of a SQL Server catalog of 100+ normalized tables. Extended the same pattern into an enrollment processing system with five sub-agents handling validation, identity checks, GPT-4o fallout classification, supplier communication drafts, and human escalation; async queues with DLQ support kept the stages isolated.
- **Email Automation Platform.** Designed an MCP-powered workflow on LangGraph, FastMCP, MongoDB, and Kafka that reconstructs email threads, classifies intent, calls retrieval tools across multiple MCP servers, and drafts grounded replies. Added resolvability classification so the system can cleanly hand off partial or ambiguous cases to a human.

Senior AI Engineer

May 2021 – Apr 2024

Bosch, Bengaluru

On the software development lifecycle (SDLC) and software quality team. Built ML tooling that plugged into defect triage, test planning, and in-cabin safety systems used across automotive product lines.

- **Ditto (bug similarity).** Semantic similarity and RAG search over an internal corpus covering years of historical defects from the SDLC pipeline. Sentence-transformer encoder for dense retrieval, Qdrant as the vector store, hybrid dense + BM25 scoring, and cross-encoder reranking on the top candidates. Wired into the internal bug tracker so triage engineers saw matching historical defects and their resolutions at the point of filing. Time-to-resolution dropped roughly 30% after rollout.
- **Interior Monitoring.** In-cabin safety models for seat-belt and drowsiness detection. Convolutional backbones for seat-belt classification and a facial-landmark + eye-state pipeline for drowsiness, trained on internal annotated driving-session data. Used GradCam++ to track down the systematic failure modes — lighting, occlusion, extreme head pose — and added evidential deep learning so the model could abstain rather than predict confidently and wrong. Error rate dropped 40% after those two changes went in.
- **Bug-to-Test Linkage.** Automated traceability between defect reports and the test cases that should have caught them. Fine-tuned an MPNet sentence encoder on internal bug-test pairs, tuned the similarity threshold against labeled data to balance precision and recall, and plugged the matcher into the test management system. This was the audit bottleneck before the system shipped; end-to-end throughput went up 75x and effective test coverage reached 95%.

Computer Vision Engineer

Oct 2020 – Apr 2021

Aimlytics, Hyderabad

- Built an automated speech dubbing pipeline that chained ASR, speaker diarization, and TTS for multilingual media workflows.
- **Indic TTS fine-tuning.** Adapted a two-stage TTS stack on the IIT Madras Indic TTS dataset: a character-level Tacotron-style acoustic model (text → mel spectrogram, ~28M params) followed by a WaveNet vocoder (spectrogram → waveform, ~24M params).
- Warm-started the acoustic model from English pretraining, reinitialized the character embedding table for Devanagari and other Indic scripts, and fine-tuned encoder and decoder on roughly 10–15 hours of studio audio per speaker. Retrained the WaveNet vocoder per speaker to pick up regional accent and timbre. The acoustic model trained on a single NVIDIA V100 at batch size 32; WaveNet needed 2–4x V100 because its autoregressive training runs at the sample level rather than the frame level.

Research Engineer

Nov 2019 – Jun 2020

Sirena Technologies, Bengaluru

- Developed an offline wake-word model using GRUs for embedded consumer devices, cutting detection latency by 13%.
- Trained a Siamese network for face recognition that hit 99.8% accuracy on the internal eval set.

Core Skills

- **GenAI and Agents:** LangChain, LangGraph, DSPy, Semantic Kernel, CrewAI, PydanticAI, MCP, A2A, Ollama
- **ML / DL:** PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face, OpenCV, Optuna
- **Training and Serving:** QLoRA, distributed data loading, optimizer-state sharding, activation checkpointing, vLLM, SGLang, TensorRT-LLM
- **Backend and Data:** FastAPI, Streamlit, Redis, MongoDB, Azure SQL, Cosmos DB, PySpark, Databricks, Snowflake, Azure Data Factory
- **Cloud and MLOps:** Azure, AWS, Vertex AI, Kubernetes (AKS), Docker, MLflow, DVC, LangSmith, Arize AI, Grafana
- **Languages:** Python, C++, Go

Research Experience

Research Fellow

Apr 2017 – Jun 2019

IIT Hyderabad, Hyderabad

- Applied deep learning research under Prof. C.V. Jawahar on road-audit systems and autonomous navigation. Most of the work was on semantic segmentation models and making them cheap enough to run in the field.
- Implemented and optimized PSPNet, ERFNet, MobileNet, and DeepLabV3 in PyTorch; several went into pilot deployments.
- Co-authored *Efficient Semantic Segmentation using Gradual Grouping*, presented at the CVPR Workshop 2018, where it picked up Best Runner-up.

Education

- **M.S. in Electrical Engineering**, Florida Institute of Technology 2016
- **B.E. in Electronics and Communication Engineering**, Manipal Institute of Technology 2014

Certifications

- *Scratch to Scale: Large-Scale Training in the Modern World* (Maven, 2025). Cohort course by Zachary Mueller covering distributed data loading, optimizer-state sharding, activation checkpointing, and multi-node training for transformer-scale models. Fine-tuned Mistral 7B on MedInstruct-52k (~45K filtered samples) with QLoRA across 4xA100s using DDP and activation checkpointing; LLM-as-judge win rate 78% vs 61% against base Mistral-7B on 500 held-out medical instructions.

Technical Reviewing

- Technical reviewer for GenAI/ML books including *AI Agents in Action*, *AI Apps with LangChain*, *Generative AI Applications with AWS*, *Optimization Bootcamp*, and *Context Engineering with DSPy*.